

# Extra-phosphate load from food additives in commonly eaten foods: a real and insidious danger for renal patients.

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**BACKGROUND:** Restriction of dietary phosphorus is a major aspect of patient care in those with renal disease. Restriction of dietary phosphorus is necessary to control for phosphate balance during both conservative therapy and dialysis treatment. The extra amount of phosphorus which is consumed as a result of phosphate-containing food additives is a real challenge for patients with renal disease and for dieticians because it represents a "hidden" phosphate load. The objective of this study was to measure phosphorus content in foods, common protein sources in particular, and comprised both those which included a listing of phosphate additives and those which did not. **METHODS:** Determinations of dry matter, nitrogen, total and soluble phosphate ions were carried out in 60 samples of foods, namely cooked ham, roast breast turkey, and roast breast chicken, of which, 30 were with declared phosphate additives and the other 30 similar items were without additives. **RESULTS:** Total phosphorus ( $290 \pm 40$  mg/100 g vs.  $185 \pm 23$  mg/100 g,  $P < .001$ ) and soluble phosphorus ( $164 \pm 25$  mg/100 g vs.  $100 \pm 19$  mg/100 g,  $P < .001$ ) content were higher in products containing additives than in foods without additives. No difference was detected between the 2 groups regarding dry matter ( $27.2 \pm 2.0$  g/100 g vs.  $26.7 \pm 1.9$  g/100 g) or total nitrogen ( $3.15 \pm 0.40$  g/100 g vs.  $3.19 \pm 0.40$  g/100 g). Consequently, phosphorus intake per gram of protein was much greater in the foods containing phosphorus additives ( $15.0 \pm 3.1$  mg/g vs.  $9.3 \pm 0.7$  mg/g,  $P < .001$ ). **CONCLUSIONS:** Our results show that those foods which contain phosphate additives have a phosphorus content nearly 70% higher than the samples which did not contain additives. This creates a special concern because this extra amount of phosphorus is almost completely absorbed by the intestinal tract. These hidden phosphates worsen phosphate balance control and increase the need for phosphate binders and related costs. Information and educational programs are essential to make patients with renal disease aware of the existence of foods with phosphate additives. Moreover, these facts highlight the need for national and international authorities to devote more attention to food labels which should clearly report the amount of natural or added phosphorus. Copyright © 2011 National Kidney Foundation, Inc. Published by Elsevier Inc. All rights reserved.